

Project 2: AI audit for health network

From Data to Solutions
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Dimitrios Dekas, Ana Sofia Luna



Agenda

1 What we have learned

2 **Motivation**
Project scenario

3 **Task**

4 **Final Hand in**

01

What you have learned

What you have learned

LIME

- **For local explanations**
- why did the model score *this one patient* the way it did

SHAP

- **For global explanations**
- what does the model rely on across hundreds of patients, on average.

Logit lens

- Takes the model's hidden state at an intermediate layer and answers what concepts lead to the decision

Sparse auto encoder

- Re-expresses a vector as a much larger but sparse set of features

02

Motivation

Future-Driven Diagnostics faces a problem



Problem

- **Slower appointment time:** patients are available months in advance
- **Slower analysis results:** Results of tests and analysis are taking longer and can put patients at risk
- **Doctors and nurses are overworked**



■ Observations made from the hospital

The hospital detected 5 of the most common reasons and procedure people seek medical attention. The hospital has accumulated seven AI systems:

- cardiovascular risk score
- hospital-readmission predictor
- stroke triage flag
- chest X-ray reader
- skin-lesion screener
- Breast-ultrasound screener
- Colon-histopathology classifier

Improvement

- **Help diagnose patients quicker:** Giving patients results quicker
- **Doctors will be able to attend more people overall**

Audit consultant

Every one of these systems reports strong accuracy. However no one has ever asked why they work. Compliance won't let any of them near a real patient until someone can answer three questions for every single system

You are hired as an AI audit consultant to review the consistency of the models to send them into production.

Model	Metric	Performance Score
Cardiovascular Risk Score	Cardiovascular risk	96.2%
Hospital-Readmission Predictor	Readmission predictor	95.8%
Stroke Triage Flag	Stroke triage flag	97.1%
Chest X-ray Reader	Chest x-ray reader	96.5%
Skin-Lesion Screener	Skin-lesion screener	95.5%
Breast-ultrasound screener	Ultrasound screener	92.0%
Colon-histopathology classifier	histopathology classifier	96.3%

- What is it using to make these decisions?
- Does this hold up or is it a shortcut that happens to correlate to the right answer?
- How do we know the explanation is trustworthy?

03

Your task

Goal

Audit report

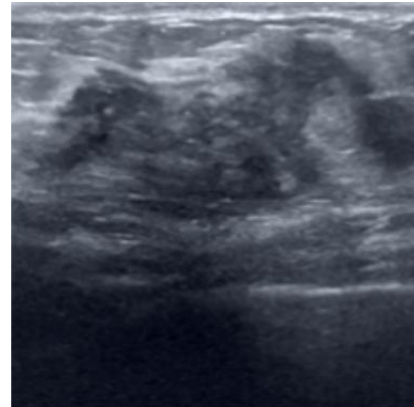
The audit loop (every system, every modality)



input



input



time in hospital	Num lab procedure	Num procedures	Num medications	Number outpatient	Number emergency	Number inpatient	Number diagnoses
0	-0.13	2.03	0.97	1.35	-0.29	-0.21	4.23
1	0.87	0.45	-0.79	1.72	-0.29	-0.21	-0.50
2	0.87	1.37	-0.20	-0.50	-0.29	-0.21	-0.50
3	0.87	-1.02	-0.20	-0.00	-0.29	-0.21	-0.50
4	2.55	1.52	2.14	2.70	-0.29	-0.21	1.08

Hand in

Audit report

- Guideline notebook for you to add your verdict
- Create a small report with your observations and fill the final scorecard as 'Trustworthy', 'Inconclusive', or 'Not ready' and answer the guideline questions

 Future-Driven Diagnostics Health -- AI Audit Scorecard			
	Cardiovascular risk score	Tabular	?
	Readmission predictor	Tabular	?
	Stroke triage flag	Tabular	?
	Chest X-ray reader	Imaging	?
	Skin-lesion screener	Imaging	?
	Breast-ultrasound screener	Imaging	?
	Colon-histopathology classifier	Imaging	?